LAB Assignment 3

Correlation Matrix

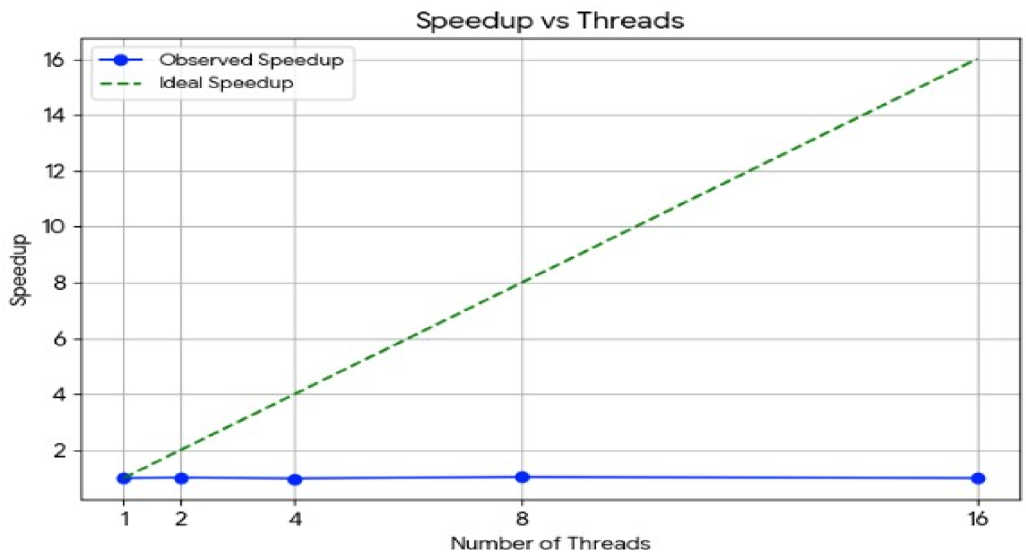
1.Sequential

Metric

Metric	Observation
Utilization	Increased up to 2 threads, reduced at 16 threads
CPI	Increased from 0.413 (2T) to 0.420 (16T)
Time (Elapsed)	Decreased from 2.037 s to 1.959 s
Page Faults	~11768 (1T) to ~13795 (Par at 16T)

Threads vs Performance

Threads (N)	Execution Time (s)	Speedup (S)	Efficiency (E=S/N)
1	2.037	1.0	100%
2	2.005	1.02	50.79%
4	2.064	0.99	24.67%
8	1.959	1.04	13.00%
16	2.038	1.00	6.25%



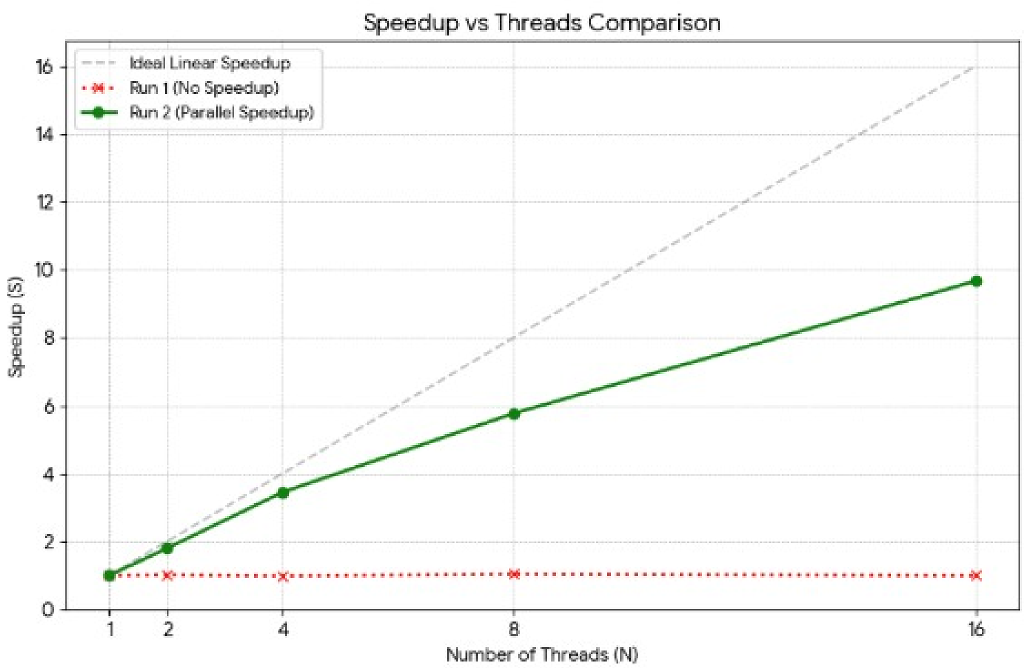
2.Parallel(OpenMp)

Metric

Metric	Observation
Utilization	Increased up to 2 threads, reduced at 16 threads
CPI	Increased from 0.234 (2T) to 0.043 (16T)
Time (Elapsed)	Decreased from 2.040 s to 0.211 s
Page Faults	~11768 (1T) to ~13795 (Par at 16T)

Threads vs Performance

Threads (N)	Execution Time (s)	Speedup (S)	Efficiency (E=S/N)
1	2.040	1.0	100%
2	1.134	1.80	89.98%
4	0.591	3.45	86.32%
8	0.354	5.77	72.08%
16	0.211	9.67	60.46%



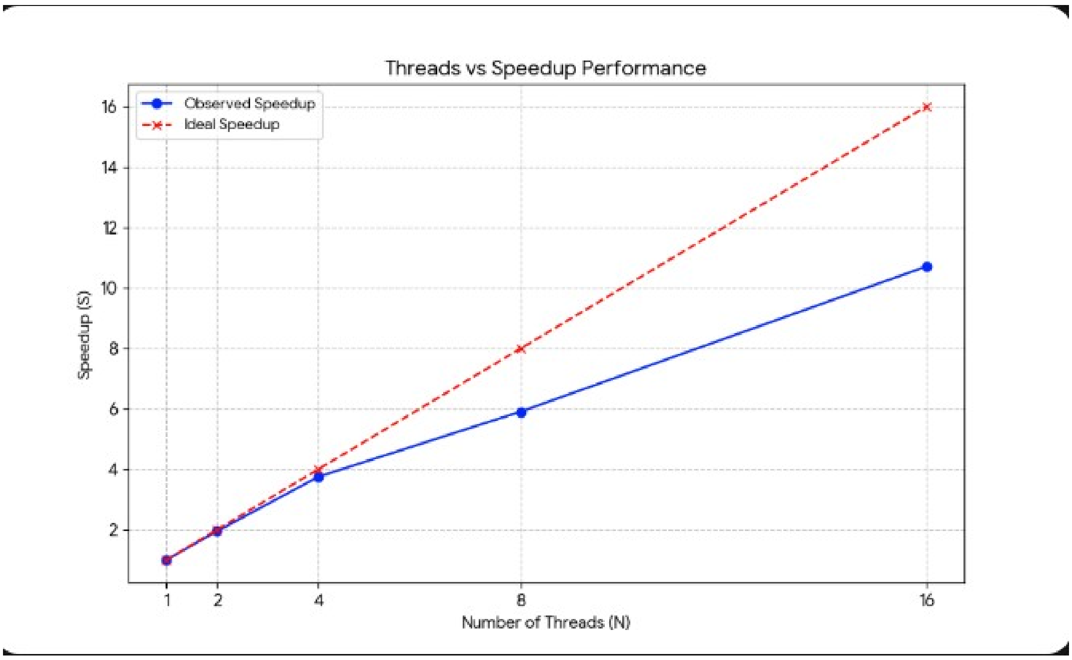
3.Optimized(SIMD+OpenMP+Cache)

Metric

Metric	Observation
Utilization	Increased up to 2 threads, reduced at 16 threads
CPI	Increased from 0.231 (2T) to 0.042 (16T)
Time (Elapsed)	Decreased from 2.182 s to 0.204 s
Page Faults	~11768 (1T) to ~13795 (Par at 16T)

Threads vs Performance

Threads (N)	Execution Time (s)	Speedup (S)	Efficiency (E=S/N)
1	2.182	1.0	100%
2	1.122	1.95	97.25%
4	0.583	3.75	93.64%
8	0.369	5.91	73.93%
16	0.204	10.71	66.95%



Conclusion

The performance comparison clearly shows that the sequential version does not scale, while the OpenMP and optimized versions achieve significant parallel speedup. In the sequential case, execution time remains almost constant around 2.037 s (1 thread) and 2.038 s (16 threads), with a maximum speedup of only 1.04 and efficiency dropping to 6.25%, proving that increasing threads provides no real benefit. In contrast, the OpenMP implementation reduces execution time from 2.040 s to 0.211 s at 16 threads, achieving a speedup of 9.67 with 60.46% efficiency, demonstrating strong parallel scalability despite some overhead. The optimized version (SIMD + OpenMP + cache) performs the best, reducing time from 2.182 s to 0.204 s, achieving the highest speedup of 10.71 and better efficiency of 66.95%.

Moreover, CPI decreases significantly in the OpenMP and optimized versions (around 0.04 at higher threads) compared to the sequential case (~0.42), indicating better instruction throughput. The optimized implementation also maintains slightly higher efficiency than OpenMP at larger thread counts, showing improved hardware utilization. Overall, combining thread-level and data-level parallelism provides the most scalable and efficient performance among the three approaches.

Version	Time @ 4T (s)	Speedup	vs Seq
Sequential	2.064	1.00x	---
OpenMP	0.591	3.49x	3.49x vs seq
SIMD+OMP+Cache	0.583	3.54x	3.54x vs seq